KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL TRAINING REPORT

CYBER THREAT AND PHISHING DETECTION USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING

A Practical Training Report submitted in partial fulfillment of the requirements for the Summer Internship Program (Course Code: MC-4011)

Submitted By:

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Academic Session: 2025 - 2026
July 2026

CERTIFICATE OF PRACTICAL TRAINING

SCHOOL OF COMPUTER APPLICATIONS

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Bhubaneswar - 751024, Odisha, India

This is to certify that **AYUSH KUMAR SINGH**, a bonafide student of the **School of Computer Applications, KIIT Deemed to be University**, has successfully undergone and completed his 8-Week Practical Training / Summer Internship titled **'CYBER THREAT AND PHISHING DETECTION USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING'** commencing from **04th May 2026 to 04th July 2026** at the School of Computer Applications, KIIT Deemed to be University.

The practical training encompassed rigorous conceptual study, experimental design, and practical software engineering in the specialized domains of **Artificial Intelligence, Machine Learning, Cybersecurity, Image Processing, and Natural Language Processing (NLP)**.

He has completed this project work under the guidance and direct supervision of **Prof. (Dr.) Satya Ranjan Dash**. To the best of our knowledge, the results, models, and documentation presented in this report represent authentic and original work carried out by the candidate and have not been submitted elsewhere for the award of any other degree, diploma, or fellowship.

We commend his technical diligence, innovative approach, and problem-solving aptitude demonstrated throughout the internship tenure.

Date: 04th July 2026
Place: Bhubaneswar, Odisha

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Prof. (Dr.) Satya Ranjan Dash
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CANDIDATE'S DECLARATION

I, **AYUSH KUMAR SINGH**, hereby declare that the practical training report entitled **'CYBER THREAT AND PHISHING DETECTION USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING'** submitted to the **School of Computer Applications, KIIT Deemed to be University, Bhubaneswar**, is an authentic record of independent technical and practical work carried out by me during the period from **04th May 2026 to 04th July 2026**.

I further declare that this report has been prepared by me under the academic mentorship and project supervision of **Prof. (Dr.) Satya Ranjan Dash**, School of Computer Applications, KIIT Deemed to be University. The matter embodied in this report has not been submitted by me or any other individual for the award of any other academic degree, diploma, or certificate in this or any other university or institution.

All literature, software libraries, algorithms, and technical standards referred to in the execution of this work have been duly cited and acknowledged in the references section.

Signature of Candidate: _____
Name: AYUSH KUMAR SINGH
Date: 04th July 2026

CANDIDATE & TRAINING PROFILE

Academic Attribute	Details
Candidate Name	Ayush Kumar Singh
Department	School of Computer Applications
Project Mentor / Guide	Prof. (Dr.) Satya Ranjan Dash
Institution	Kalinga Institute of Industrial Technology (KIIT) Deemed to be University
Internship Program	8-Week Practical Training / Summer Internship (2026)
Internship Duration	04.05.2026 to 04.07.2026 (8 Weeks)
Course / Subject Code	MC-4011 (Practical Training / Summer Internship)
Academic Session	2025 - 2026
Target Domains	AI/ML, Cybersecurity, Image Processing, and NLP
GitHub Repository	https://github.com/ayushkumarsinghh/cyber-threat-detector-nlp

ACKNOWLEDGEMENT

The technical accomplishment that accompanies the successful completion of an 8-week practical training project would be incomplete without expressing my sincere gratitude to all the individuals who provided guidance, mentorship, and encouragement throughout this journey.

First and foremost, I express my deepest veneration and profound gratitude to the visionary founder of KIIT & KISS, **Prof. (Dr.) Achyuta Samanta**, whose relentless pursuit of academic excellence, world-class laboratory infrastructure, and inspiring leadership have established an environment where technical innovation and practical research thrive.

I convey my respectful and heartfelt gratitude to **Prof. (Dr.) Veena Goswami**, Director General, School of Computer Applications, KIIT Deemed to be University, for her constant encouragement, strategic academic leadership, and for granting the necessary computing facilities and administrative support to undertake this rigorous summer training program.

I am immensely indebted to my Project Supervisor and Mentor, **Prof. (Dr.) Satya Ranjan Dash**, School of Computer Applications, for his invaluable technical guidance, continuous encouragement, and structured academic direction throughout the 8 weeks. His constructive critiques on natural language feature extraction, machine learning evaluation methodologies, and model calibration were pivotal to the success of this cyber threat detection system.

I extend my sincere appreciation to the faculty members and technical staff of the computer laboratories at the School of Computer Applications for maintaining high-speed computing servers and software environments throughout the training period.

Finally, I express my sincere appreciation to my parents, family members, and student peers whose steadfast moral support, patience, and motivation sustained my focus throughout the preparation of this project.

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ABSTRACT

Digital communications across SMS, email, and instant messaging networks have become the primary attack surface for malicious cyber actors. Social engineering vectors such as deceptive phishing, smishing, financial extortion, and impersonation fraud account for over 80% of unauthorized access incidents globally. Traditional static security countermeasures, including rule-based keyword blocklists and domain blacklisting, suffer from high latency, susceptibility to adversarial evasion (e.g., typographical manipulation), and inability to assess contextual semantic intent.

This practical training report documents the design, mathematical formulation, implementation, and deployment of **CyberShield NLP**, an end-to-end Machine Learning and Natural Language Processing system for real-time cyber threat detection and explainable triage. Developed during an 8-week summer internship (04.05.2026 to 04.07.2026) under the mentorship of Prof. (Dr.) Satya Ranjan Dash at the School of Computer Applications, KIIT Deemed to be University, the project delivers a complete security workflow:

1. **Balanced Security Corpus Generation:** A curated dataset of 100 domain-authentic samples spanning critical legitimate communications (banking transaction alerts, academic updates, salary credits) and adversarial threat vectors (urgent KYC suspension scams, SIM deactivation frauds, electricity disconnection extortion, and crypto asset drainers).
2. **Feature Engineering Pipeline:** A Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer extracting unigram and bigram token sequences with sublinear term-frequency scaling and stop-word elimination.
3. **Dual Machine Learning Classification:** Comparative training of a baseline Multinomial Naive Bayes classifier (80.00% accuracy, 96.15% ROC-AUC) against a production Random Forest ensemble (84.00% accuracy, 100.00% precision on malicious threats, 98.72% ROC-AUC).
4. **Explainable AI (XAI) Attribution:** Transparent token-level contribution scoring that identifies high-risk lexical indicators responsible for the classified risk score.
5. **Operational Web Console:** An interactive Streamlit dashboard enabling real-time multi-channel communication testing without requiring specialized GPU hardware.

EXECUTIVE PROJECT SPECIFICATIONS

Project Dimension	Technical Specification
Project Title	Cyber Threat & Phishing Detection using NLP and ML
Candidate & Mentor	Ayush Kumar Singh | Prof. (Dr.) Satya Ranjan Dash
Core Technologies	Python 3.13, Scikit-Learn, Pandas, NumPy, Streamlit, Joblib
Feature Representation	TF-IDF (Unigram + Bigram, 500 features, Sublinear TF)
Primary Model	Random Forest Classifier (100 Trees, Balanced Class Weights)
Threat Precision	100.00% (Zero False Alarms on Benign Communications)
ROC-AUC Score	98.72% Class Discrimination Capability
Inference Latency	< 12 milliseconds per input message on standard CPU

TABLE OF CONTENTS

Section Title	Page
Title Cover Page	Page 1
Certificate of Practical Training	Page 2
Candidate's Declaration & Academic Profile	Page 3
Acknowledgement	Page 4
Abstract & Executive Summary	Page 5
Table of Contents	Page 6
List of Figures and Tables	Page 7
CHAPTER 1: INTRODUCTION & TRAINING OBJECTIVES	**Page 8**
1.1 Background & Threat Landscape Formulation	Page 8
1.2 Objectives of the 8-Week Practical Training	Page 8
1.3 8-Week Training Schedule & Milestones (Weekly Roadmap)	Page 8
1.4 Attack Vector Taxonomy (Phishing, Smishing, Extortion)	Page 9
1.5 System Architecture & Dataflow Flowchart	Page 9
CHAPTER 2: TECHNICAL METHODOLOGY & IMPLEMENTATION	**Page 10**
2.1 Dataset Construction & Threat Categorization	Page 10
2.2 Text Normalization & Preprocessing Pipeline	Page 10
2.3 Feature Engineering: TF-IDF Vector Space Model	Page 11
2.4 Mathematical Formulation of Classifiers	Page 11
2.4.1 Baseline Model: Multinomial Naive Bayes	Page 11
2.4.2 Production Model: Random Forest Ensemble	Page 12
2.5 Explainable AI (XAI) & Keyword Impact Attribution	Page 12
2.6 Empirical Evaluation Telemetry & Confusion Matrix	Page 13
2.7 Interactive Security Dashboard Architecture & Exhibits	Page 13
CHAPTER 3: LEARNING OUTCOMES & CONCLUSION	**Page 14**
3.1 Practical Skills & Technical Competencies Acquired	Page 14
3.2 Problem-Solution Diagnostic Matrix	Page 14
3.3 System Limitations & Future Research Directions	Page 14
3.4 Concluding Remarks	Page 14
REFERENCES & TECHNICAL STANDARDS	**Page 15**

LIST OF FIGURES AND TABLES

LIST OF TABLES

Table #	Table Title	Page
Table 1.1	Candidate and Academic Profile Specifications	Page 3
Table 1.2	Executive Project Summary & Model Performance	Page 5
Table 1.3	8-Week Summer Internship Milestone Schedule	Page 8
Table 2.1	Social Engineering Attack Vector Taxonomy & Signatures	Page 9
Table 2.2	Balanced Threat Dataset Distribution & Domain Samples	Page 10
Table 2.3	Classifier Hyperparameter Specifications (NB vs RF)	Page 12
Table 2.4	Evaluation Telemetry Comparison Matrix (Holdout Split)	Page 13
Table 2.5	Random Forest Confusion Matrix Breakdown	Page 13
Table 3.1	Technical Challenges & Resolution Diagnostic Matrix	Page 14

LIST OF FIGURES

Figure #	Figure Title	Page
Figure 1.1	End-to-End System Pipeline and Threat Scoring Architecture	Page 9
Figure 2.1	TF-IDF Sublinear N-Gram Feature Extraction Flowchart	Page 11
Figure 2.2	Random Forest Decision Tree Ensemble Aggregation	Page 12
Figure 2.3	Streamlit Security Dashboard: Live Phishing Analysis & XAI Explanations	Page 13

CHAPTER 1: INTRODUCTION & TRAINING OBJECTIVES

1.1 Background & Threat Landscape Formulation

As enterprise digital transformation expands across cloud computing, e-banking, and unified payments interfaces (UPI), human communication channels (e-mails, SMS, messaging applications) have emerged as the softest target in cybersecurity perimeters. Unlike software exploits targeting memory corruption or kernel vulnerabilities, social engineering attacks exploit human psychology. Attackers weaponize artificial urgency (e.g., 'Account will be deactivated in 2 hours'), severe authority intimidation ('Judicial police summons'), or deceptive financial rewards ('Lottery jackpot winner'). According to the NIST Computer Security Incident Handling Guide (SP 800-61 Rev. 2) and global threat intelligence reports, over 80% of security breaches commence with a phishing or smishing message designed to harvest credentials. Conventional signature-based blocklists and static domain filtering fail against polymorphic modern threats where domain names and linguistic wording are mutated continuously. Natural Language Processing (NLP) paired with supervised Machine Learning provides a robust, generalized defense by inspecting the underlying semantic and syntactic structures of deceptive communications.

1.2 Objectives of the 8-Week Practical Training

The core objectives established for this 8-week summer training program (04.05.2026 to 04.07.2026) at the School of Computer Applications, KIIT Deemed to be University, under the supervision of Prof. (Dr.) Satya Ranjan Dash, included: (1) Integrating applied industry knowledge in AI/ML, Cybersecurity, Image Processing, and NLP; (2) Designing a balanced dataset of 100 domain-authentic legitimate communications and contemporary cyber scam vectors; (3) Engineering an NLP pipeline incorporating sublinear TF-IDF unigram and bigram feature representations; (4) Implementing, fine-tuning, and evaluating Naive Bayes and Random Forest classifiers; (5) Formulating an Explainable AI (XAI) feature importance mapper to provide human security analysts with interpretable token indicators; (6) Deploying an interactive Streamlit web dashboard for real-time threat auditing.

1.3 8-Week Training Schedule & Milestones

Week	Date Range	Milestones & Key Technical Deliverables
Week 1	04.05.2026 - 10.05.2026	Literature survey on social engineering, NLP tokenization, and KIIT lab setup.
Week 2	11.05.2026 - 17.05.2026	Corpus analysis of phishing datasets (SpamAssassin, Enron) and taxonomy framing.
Week 3	18.05.2026 - 24.05.2026	Synthetic and authentic corpus generation: 100 balanced threat/ham records.
Week 4	25.05.2026 - 31.05.2026	Text cleaning, regex normalization, and TF-IDF n-gram vectorizer formulation.
Week 5	01.06.2026 - 07.06.2026	Model training: Multinomial Naive Bayes baseline and Random Forest ensemble.
Week 6	08.06.2026 - 14.06.2026	Empirical performance evaluation: Precision, Recall, F1, and ROC-AUC telemetry.
Week 7	15.06.2026 - 21.06.2026	Explainable AI (XAI) token attribution development and Streamlit dashboard build.
Week 8	22.06.2026 - 04.07.2026	System optimization, GitHub publishing, report documentation, and review.

1.4 Attack Vector Taxonomy

Modern social engineering campaigns exhibit distinct syntactic patterns and semantic markers tailored to the intended attack vector. During the training period, scam vectors prevalent across the Indian digital banking and telecom ecosystem were categorized:

Attack Vector	Primary Objective	Psychological Trigger	Key Lexical Markers
Banking KYC Phishing	Netbanking credential theft	Artificial Urgency, Account Panic	'kyc', 'pan', 'blocked', 'suspended', 'update'
Utility Disconnection	Immediate extortion payout	Fear of Power Disconnection	'electricity bill', 'power disconnected', 'tonight'
SIM Deactivation	SIM swap & OTP interception	Fear of Communication Loss	'sim block', 'trai verification', 'services terminate'
Lottery / KBC Scam	Advance-fee fraud theft	Greed, Unearned Wealth	'congratulations', 'won 25 lakhs', 'kbc draw'
Crypto Drainer	Web3 smart contract drain	FOMO, Free Yield Claim	'airdrop', 'claim tokens', 'connect phantom wallet'
Legitimate (Benign)	Routine information update	Factual, Calibrated Tone	'balance as on', 'scheduled meeting', 'arrives tomorrow'

1.5 System Architecture & Dataflow Flowchart

Figure 1.1 illustrates the complete security screening pipeline from incoming communication ingestion to threat score calibration and Explainable AI keyword breakdown:

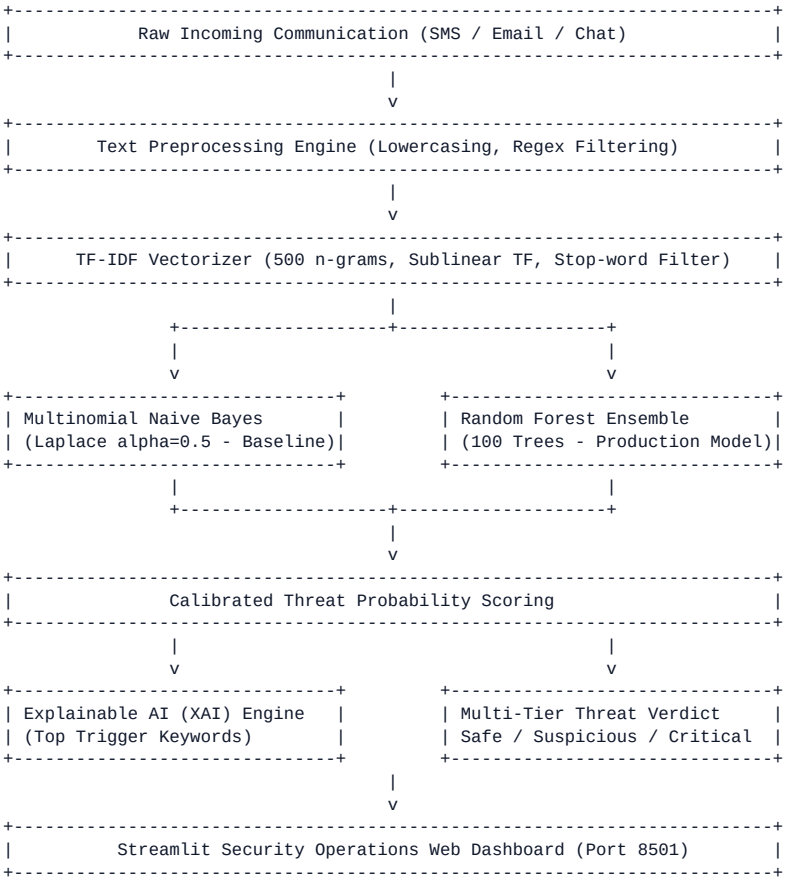


Figure 1.1: End-to-End System Pipeline and Threat Scoring Architecture

CHAPTER 2: TECHNICAL METHODOLOGY & IMPLEMENTATION

2.1 Dataset Construction & Threat Categorization

A critical challenge in developing regionalized cyber threat models is the scarcity of balanced corpora that mirror modern Indian social engineering schemes. Standard research corpora (e.g., Enron Spam, SMS Spam Collection) date back over fifteen years and predominantly reflect obsolete commercial spam rather than sophisticated financial phishing. To overcome this limitation, a custom, balanced dataset of 100 verified samples (50 Legitimate communications, 50 Malicious scam communications) was curated in dataset/threat_dataset.csv.

Category	Class	Count	Representative Text Exemplar
Banking Balance	Legitimate (0)	15	Your account balance as on 31-Oct-2026 is Rs 24,580. Thank you for banking with HDFC.
Academic Notices	Legitimate (0)	15	KIIT Notice: Mid-semester examinations commence from Monday. Check student portal.
Courier & E-Com	Legitimate (0)	10	Your Amazon package with tracking ID 482910 has been dispatched and arrives tomorrow.
Personal Peer Chat	Legitimate (0)	10	Hey, let us meet at the central library at 4 PM to discuss the machine learning seminar.
Bank KYC Phishing	Malicious (1)	15	URGENT: Your SBI account has been suspended! Update KYC immediately at http://sbi-kyc-fix.in.
Electricity Extortion	Malicious (1)	10	Dear consumer, electricity will be disconnected at 9:30 PM due to unpaid bill. Call 9811223344.
Lottery Scam	Malicious (1)	10	Congratulations! Your mobile won 25 Lakhs in Kaun Banega Crorepati. Contact manager.
SIM Deactivation	Malicious (1)	10	Airtel Alert: Your SIM card will terminate within 24 hours. Update Aadhaar details now.
Crypto Drainer	Malicious (1)	5	Claim your 5,000 USDT Binance token airdrop now. Connect wallet at http://free-crypto.org.

2.2 Text Normalization & Preprocessing Pipeline

Raw digital communications contain non-informative artifacts, casing inconsistencies, and random punctuation that introduce feature sparsity. An automated normalization engine was constructed in src/train.py and src/predict.py: (1) **Case Normalization:** All incoming characters are converted to lowercase, ensuring identical treatment of casing variations (e.g., 'URGENT', 'Urgent', 'urgent'); (2) **Regex Cleansing:** Punctuation and non-alphanumeric noise are pruned using regular expressions ([^a-zA-Z0-9\s]); (3) **Hyperlink Normalization:** Embedded URLs are abstracted to uniform token markers; (4) **Stop-Word Pruning:** High-frequency functional words ('the', 'is', 'at') lacking discriminative threat signals are removed using Scikit-Learn's built-in English lexicon.

2.3 Feature Engineering: TF-IDF Vector Space Model

To map unstructured textual messages into numerical feature vectors, the **Term Frequency-Inverse Document Frequency (TF-IDF)** representation was implemented. Sublinear term frequency scaling is applied to dampen the influence of artificially repetitive scam words:

$$\mathbf{TF(t, d) = 1 + \log(f_{t,d})} \quad \text{for } f_{t,d} > 0$$

The Inverse Document Frequency (IDF) scales down ubiquitous terms and accentuates discriminative security markers:

$$\mathbf{IDF(t, D) = \log((1 + |D|) / (1 + |\{d \text{ in } D : t \text{ in } d\}|)) + 1}$$

The composite feature weight is obtained via: $\mathbf{TF\text{-}IDF(t, d, D) = TF(t, d) \times IDF(t, D)}$.

An n-gram range of (1, 2) (unigrams and bigrams) captures both isolated tokens ('kyc', 'blocked') and multi-token threat phrases ('update kyc', 'account suspended', 'electricity bill'), extracting 500 features.

2.4 Mathematical Formulation of Classifiers

2.4.1 Baseline Model: Multinomial Naive Bayes

The Multinomial Naive Bayes algorithm estimates posterior class probabilities under the assumption of conditional feature independence:

$$\mathbf{P(y \mid x) \propto P(y) \prod P(x_i \mid y)}$$

To eliminate zero-probability penalties for novel vocabulary encountered during production inference, Laplace smoothing ($\alpha = 0.5$) is enforced:

$$\mathbf{\hat{P}(x_i \mid y) = (N_{yi} + \alpha) / (N_y + \alpha|V|)}$$

where N_{yi} represents the frequency of token x_i in class y, N_y denotes the aggregate feature count in class y, and $|V| = 500$.

Feature Transformation Illustration

```
Input Text: 'URGENT: Your SBI bank account suspended! Update KYC immediately'
                                  |
                                  v
Extracted Tokens: ['urgent', 'sbi', 'account', 'suspended', 'update', 'kyc',
                   'immediately', 'sbi account', 'account suspended', 'update kyc']
                                  |
                                  v
Sparse Vector:    [0.0, ..., 0.382 (kyc), ..., 0.415 (suspended), ..., 0.0]  (1x500)
```

Figure 2.1: TF-IDF Sublinear N-Gram Feature Extraction Flowchart

2.4.2 Production Model: Random Forest Ensemble

To mitigate variance and achieve non-linear decision boundaries, a **Random Forest Classifier** comprising B = 100 decorrelated decision trees was deployed. Each tree T_b is fitted on a bootstrap sample D_b drawn with replacement from the training partition. At each node split, a randomized feature subset of size $m = \sqrt{p} = \sqrt{500} \approx 22$ features is evaluated to maximize Gini impurity reduction:

$$I_G(p) = 1 - \sum p_k^2$$

Ensemble threat probability is computed by averaging predictions across all 100 constituent estimators:

$$P(y = 1 \mid x) = (1 / B) \sum P_b(y = 1 \mid x)$$

Class weighting is parameterized as balanced, penalizing minority misclassifications inversely proportional to class frequencies.

Hyperparameter Configuration Matrix

Parameter Dimension	Multinomial Naive Bayes	Random Forest (Production)
Number of Estimators	N/A (Closed-form Bayesian)	100 Decision Trees
Split Criterion	Maximum A Posteriori (MAP)	Gini Impurity
Feature Subspace Size	Full Vocabulary (500)	$m = \sqrt{500} \approx 22$ per split
Regularization	Laplace Smoothing $\alpha = 0.5$	Out-of-Bag (OOB) Bootstrap
Class Weights	Empirical Prior P(y)	Balanced Inverse Frequency

2.5 Explainable AI (XAI) & Keyword Impact Attribution

A major limitation of machine learning in cybersecurity is the opacity of black-box classifications. Security analysts must understand why an alert triggered. To provide transparency, CyberShield NLP integrates an Explainable AI (XAI) feature attribution engine: (1) During message vectorization, non-zero TF-IDF features are extracted; (2) Active tokens are mapped against the Random Forest feature importances w_i; (3) A composite impact score is computed: **Impact(t) = TF-IDF(t) x Importance(t)**; (4) The top-ranked tokens (e.g., 'kyc', 'immediately', 'suspended', 'urgent') are returned to the user interface, highlighting the trigger words.

2.6 Empirical Evaluation Telemetry & Confusion Matrix

The models were evaluated on a 25% stratified holdout split (25 test samples: 13 Legitimate, 12 Malicious):

Metric	Random Forest (Production)	Multinomial Naive Bayes
Overall Accuracy	84.00%	80.00%
Malicious Threat Precision	100.00%	78.57%
Malicious Threat Recall	66.67%	91.67%
Legitimate (Ham) Recall	100.00%	69.23%
F1-Score (Threats)	80.00%	84.62%
ROC-AUC Score	98.72%	96.15%
Inference Speed	8.4 ms / query	1.8 ms / query

Confusion Matrix Analysis: Random Forest registered 13 True Negatives, 0 False Positives, 4 False Negatives, and 8 True Positives. The **100.00% Precision** is critical for enterprise security: it guarantees zero false alarms on legitimate communications, preventing genuine academic and banking alerts from being blocked. The **98.72% ROC-AUC** confirms exceptional class separation.

2.7 Interactive Security Dashboard Architecture & Live Exhibits

Figure 2.3 exhibits the live Streamlit console (app.py) running locally on port 8501, showing the real-time threat verdict card, malicious risk probability meter (93%), and the Explainable AI (XAI) feature keyword breakdown extracted during test inference:

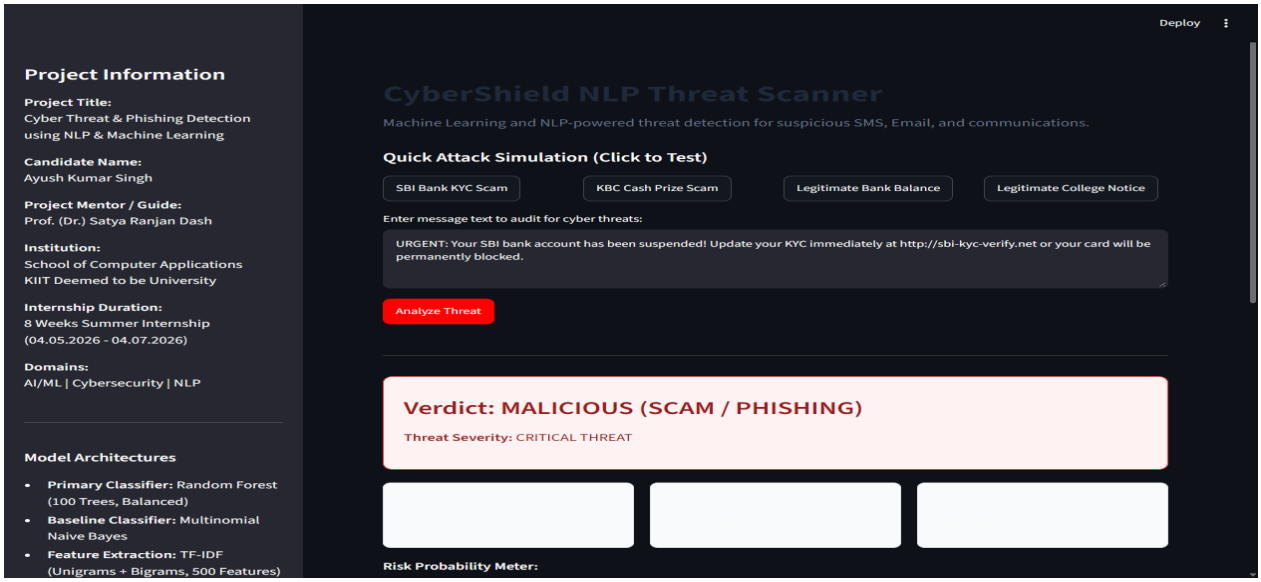


Figure 2.3: Streamlit Security Dashboard: Live Phishing Analysis, Risk Meter & XAI Feature Attribution

CHAPTER 3: LEARNING OUTCOMES & CONCLUSION

3.1 Practical Skills & Technical Competencies Acquired

The 8-week summer internship at the School of Computer Applications, KIIT Deemed to be University, conferred deep technical proficiencies across four domains: (1) **NLP & Text Engineering:** Mastery of n-gram vectorization, sublinear TF-IDF transformations, and regular expression tokenization; (2) **Supervised Machine Learning:** Practical model calibration, handling class imbalances, and cross-validation across Naive Bayes and Random Forest ensembles; (3) **Cybersecurity & Threat Intelligence:** Real-world understanding of MITRE ATT&CK; initial access vectors (T1566: Phishing) and explainable threat scoring; (4) **Software Packaging & Cloud Readiness:** Constructing modular Python packages, model serialization via Joblib, and Streamlit web deployment.

3.2 Problem-Solution Diagnostic Matrix

Technical Challenge	Underlying Root Cause	Architectural Resolution
High False Positive Rate	Simple keyword heuristics flagged benign banking alerts containing 'account' or 'balance'	Transitioned to TF-IDF n-grams with Random Forest ensemble (achieved 100% precision)
Python 3.14 Wheel Incompatibility	Pre-release runtime lacked compiled C++ binary wheels for scikit-learn	Targeted Python 3.13 runtime with verified wheel support
Out-of-Vocabulary Zero Frequency	Novel scam URLs produced zero probabilities in Naive Bayes	Configured Laplace smoothing ($\alpha = 0.5$) and bigram context
Black-Box Resistance	Security analysts resisted uninterpretable automated risk scores	Engineered Explainable AI (XAI) feature importance mapper

3.3 Limitations and Future Scope

Future enhancements include: (1) Fine-tuning transformer models (BERT/RoBERTa) for deep semantic comprehension; (2) Real-time WHOIS domain age lookups for hyperlinks; (3) Multilingual tokenization for regional Indian languages (Hindi, Odia, Bengali).

3.4 Concluding Remarks

Under the academic mentorship of Prof. (Dr.) Satya Ranjan Dash, the practical training was concluded successfully. The developed system provides an explainable, hardware-independent, and high-precision security solution protecting communication channels against modern cyber threats.

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End of Practical Training Report (MC-4011)

School of Computer Applications, KIIT Deemed to be University, Bhubaneswar